MATHS MANTHAN
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Subject: Mathematics **PERIODIC ASSESSMENT TEST – 07 (LINEAR EQUATIONS)** **Time:** 60 Minutes

Class: Class X (CBSE) **Chapter 3: Pair of Linear Equations in Two Variables** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 14(iv).
2. **Sec A:** 4 MCQs (1M). **Sec B:** 2 AR (1M). **Sec C:** 3 VSAQ (2M). **Sec D:** 3 SA (3M). **Sec E:** 1 LA (5M). **Sec F:** 1 Case Study (4M).

SECTION A: Multiple Choice Questions (4 × 1 = 4 Marks)

Q1. The area of the triangle formed by the lines $x = 3, y = 4$ and $x = y$ is: [1]
(A) 1/2 sq. unit (B) 1 sq. unit (C) 2 sq. units (D) None of these

Q2. Which of the following is represented by the system $3x + 4y = 5$ and $6x + 8y = 7$? [1]
(A) Parallel lines (B) Intersecting lines (C) Coincident lines (D) Perpendicular lines

Q3. A linear equation in two variables $ax + by + c = 0$ (a, b not both zero) graphically represents: [1]
(A) a circle (B) a straight line (C) a parabola (D) a hyperbola

Q4. For what value of k does the system $8x + 5y = 9$ and $kx + 10y = 15$ have no solution? [1]
(A) 8 (B) 16 (C) 4 (D) 20

SECTION B: Assertion-Reason Questions (2 × 1 = 2 Marks)

Directions: (A) Both (A) and (R) are true, and (R) is correct explanation. (B) Both true, but (R) is not correct explanation. (C) (A) is true, (R) is false. (D) (A) is false, (R) is true.

Q5. **Assertion (A):** The linear equations $x - 2y = 0$ and $3x + 4y - 20 = 0$ intersect at $(4, 2)$.
Reason (R): A point lies on both lines if and only if its coordinates simultaneously satisfy both equations. [1]

Q6. **Assertion (A):** The line represented by $y - 3 = 0$ is parallel to the x-axis.
Reason (R): Any equation of the form $y = c$ ($c \neq 0$) represents a straight line parallel to the x-axis. [1]

SECTION C: Very Short Answer Questions (3 × 2 = 6 Marks)

Q7. Solve for x and y: $\sqrt{2}x + \sqrt{3}y = 0$ and $\sqrt{3}x - \sqrt{8}y = 0$. [2]

Q8. Find the value of c for which the system $cx + 3y = 3$ and $12x + cy = 6$ has no solution. [2]

Q9. Half the perimeter of a rectangular garden, whose length is 4 m more than its width, is 36 m. Find the dimensions of the garden. [2]

SECTION D: Short Answer Questions (3 × 3 = 9 Marks)

Q10. A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Saritha paid Rs. 27 for a book kept for seven days, while Susy paid Rs. 21 for the book she kept for five days. Find the fixed charge and the charge for each extra day. [3]

Q11. Meena went to a bank to withdraw Rs. 2000. She asked the cashier to give her Rs. 50 and Rs. 100 notes only. Meena received 25 notes in all. Find how many notes of Rs. 50 and Rs. 100 she received. [3]

Q12. Solve for x and y: $\frac{x}{a} + \frac{y}{b} = 2$ and $ax - by = a^2 - b^2$. [3]

SECTION E: Long Answer / HOTS Question (1 × 5 = 5 Marks)

Q13. Two years ago, a father was five times as old as his son. Two years later, his age will be 8 years more than three times the age of the son. Formulate the pair of linear equations and find their present ages. [5]

SECTION F: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q14. Case Study: Public Municipal Library Book Lending System

A city central library promotes reading by operating an affordable lending window. The borrowing rules specify a fixed fee of Rs. x for the first 3 days and an additional fee of Rs. y for each day thereafter. Radhika borrowed a science encyclopedia for 7 days and paid Rs. 27. Aakash borrowed a history reference manual for 5 days and paid Rs. 21.

Based on the above lending scheme, answer the following questions:

(i) Formulate the pair of linear equations in x and y representing Radhika's and Aakash's borrowings. [1]

(ii) Determine the fixed charge for the first three days (x). [1]

(iii) Determine the additional daily charge per extra day (y). [1]

(iv) If Priya borrows a novel for 10 days, calculate the total amount she will have to pay. [1]

OR

If Aman paid an amount of Rs. 45, find how many days he kept the borrowed book. [1]

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